

BLUESTOCK MUTUAL FUND CAPSTONE REPORT

End-to-End Data Engineering Pipeline, Analytics Warehouse & Recommender Engine

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Repository: <https://github.com/Arushi3154/mutual-fund-etl>

1. Executive Summary

This report presents the technical design and quantitative findings of the Bluestock Mutual Fund Capstone project. The project addresses data fragmentation across Indian asset management companies by constructing an automated ETL pipeline, a normalized SQLite relational warehouse (bluestock_mf.db), a quantitative risk engine, and an interactive PowerBI analytics dashboard.

2. Data Sources & Ingestion Layer

Data was acquired from official AMFI endpoints (mfapi.in) and structured historical dataset archives. Ingested tables include Scheme Master metadata, daily NAV time-series, benchmark indices (Nifty 50), category inflows, and portfolio holdings.

3. ETL Architecture & Data Cleaning

The ETL architecture standardizes date formats, eliminates split/merger anomalies, and imputes weekend/holiday NAV gaps using forward fills (ffill()). CAGR is accurately annualized using trading days (252 days/year) rather than calendar days.

Metric	Mathematical Formula	Key Parameter / Description
CAGR	$(NAV_end / NAV_start)^{(252 / n_trading_days)} - 1$	Trading-day annualized growth rate.
Sharpe Ratio	$(R_p - R_f) / StDev(R_p)$	Risk-free rate (Rf) set to 6.5%.
Beta	$Covariance(R_p, R_m) / Variance(R_m)$	Benchmark (Rm) = Nifty 50 Index.
95% VaR	Percentile(Daily Returns, 5)	Quantifies 1-day maximum expected loss.

4. Exploratory Data Analysis (EDA) Findings

Analysis reveals that equity schemes represent over 60% of overall industry AUM. Direct plans consistently generate higher CAGR compared to regular plans due to lower expense ratios (0.8% vs 1.8%).

5. Performance & Risk Analytics

Top-performing schemes in the Large-Cap and Flexi-Cap categories exhibit Sharpe Ratios exceeding 1.25 with Betas ranging between 0.85 and 1.05, indicating strong risk-adjusted alpha generation.

6. Interactive PowerBI Dashboard Architecture

The PowerBI dashboard (bluestock_mf.pbix) comprises 4 interactive pages: (1) Executive KPI Overview, (2) Category Inflow Trends, (3) Risk vs Return Matrix, and (4) Fund Recommendation Scorecard. Every page includes at least two functional slicers (AMC, Category, Date Range).

7. Limitations & Strategic Recommendations

Limitations: Historical NAV backtesting does not fully account for extreme macroeconomic black-swan events or illiquidity in small-cap assets during market downturns.

Recommendations: Investors should rebalance portfolios quarterly targeting funds with Sharpe > 1.2 and positive CAPM Alpha.